Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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SCAN THE QR CODE
youtube.com/@hannahkettlemaths

Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 9

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1a	Solving inequalities with the variable on both sides	2	U738
1b	Solving double inequalities	3	U145
3i	Constructing fractions, Finding fractions of amounts	2	U163,U916
3ii	Selecting sampling techniques	1	U162
5	Finding unknown sides in right-angled triangles	2	U283
7	Sharing amounts in a given ratio	4	U577
9a	Similar shapes, Converting between ratios, fractions and percentages	2	U630,U176
9b	Similar shapes, Converting between ratios, fractions and percentages	2	U630,U176
11a	Drawing cumulative frequency graphs	1	U182
11b	Drawing cumulative frequency graphs	2	U182
11c	Interpreting cumulative frequency graphs	3	U642
13	Dividing algebraic fractions	4	U824
15	Changing the subjects of formulae	3	U556
17a	Sharing amounts in a given ratio, Finding the lowest common multiple (LCM)	1	U577,U751
17b	Ratios, Finding the lowest common multiple (LCM)	3	U753,U577,U751
19	Using Pythagoras' theorem in 3D, Trigonometry in 3D shapes	4	U541,U170
	Total	39	

Answer ALL questions.

Write your answers in the spaces provided.

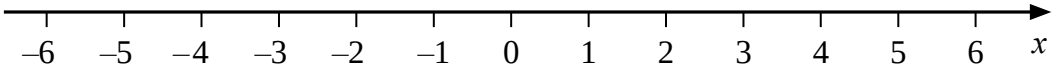
You must write down all the stages in your working.

1 (*a*) Solve $14n > 11n + 6$

.....

(2)

(*b*) On the number line below, show the set of values of x for which $-2 < x + 3 \leqslant 4$



(3)

(Total for Question 1 is 5 marks)

3 Hannah is planning a day trip for 195 students.

She asks a sample of 30 students where they want to go.
Each student chooses one place.

The table shows information about her results.

Place	Number of students
Theme Park	10
Theatre	5
Sports Centre	8
Seaside	7

(i) Work out how many of the 195 students you think will want to go to the Theme Park.

.....

(2)

(ii) State any assumption you made **and** explain how this may affect your answer.

.....

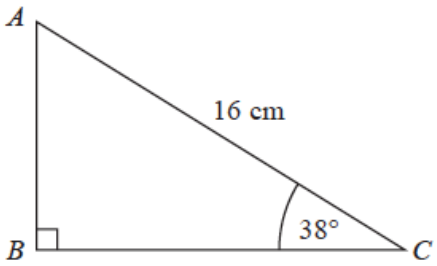
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(1)

(Total for Question 3 is 3 marks)

5 *ABC* is a right-angled triangle.



Calculate the length of *AB*.
Give your answer correct to 2 decimal places.

.....cm

(Total for Question 5 is 2 marks)

---

7 £360 is shared between Abby, Ben, Chloe and Denesh.

The ratio of the amount Abby gets to the amount Ben gets is 2 : 7

Chloe and Denesh each get 1.5 times the amount Abby gets.

Work out the amount of money that Ben gets.

£.....

(Total for Question 7 is 4 marks)

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9 The circumference of circle **B** is 90% of the circumference of circle **A**.

(*a*) Find the ratio of the area of circle **A** to the area of circle **B**.

.....

(2)

Square **E** has sides of length *e* cm.
Square **F** has sides of length *f* cm.

The area of square **E** is 44% greater than the area of square **F**.

(*b*) Work out the ratio *e* : *f*

.....

(2)

(Total for Question 9 is 4 marks)

11 The grouped frequency table gives information about the times, in minutes, that 80 office workers take to get to work.

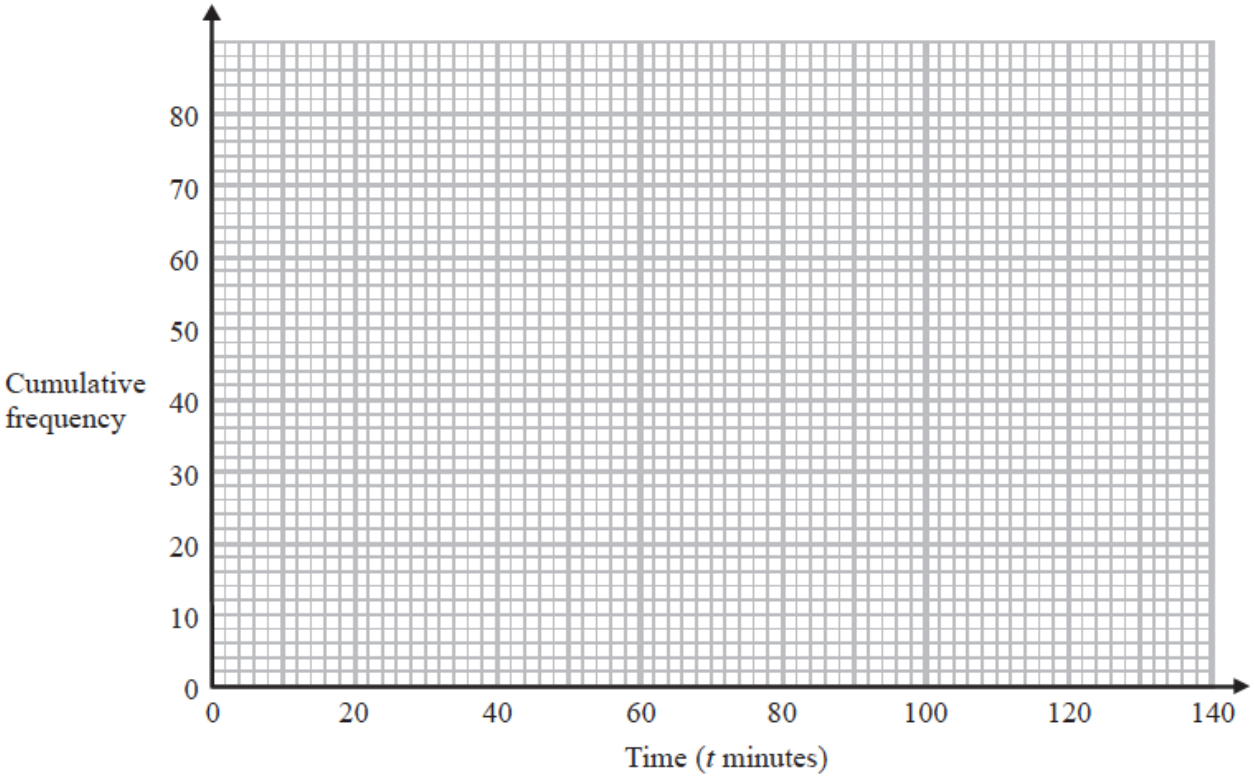
Time (*t* minutes)	**Frequency**
$0 < t \leqslant 20$	5
$20 < t \leqslant 40$	30
$40 < t \leqslant 60$	20
$60 < t \leqslant 80$	15
$80 < t \leqslant 100$	8
$100 < t \leqslant 120$	2

(*a*) Complete the cumulative frequency table.

Time (*t* minutes)	**Cumulative frequency**
$0 < t \leqslant 20$	
$0 < t \leqslant 40$	
$0 < t \leqslant 60$	
$0 < t \leqslant 80$	
$0 < t \leqslant 100$	
$0 < t \leqslant 120$	

(1)

(*b*) On the grid, draw the cumulative frequency graph for this information.



(2)

(*c*) Use your graph to find an estimate for the percentage of these office workers who take more than 90 minutes to get to work.

.....%

(3)

(Total for Question 11 is 6 marks)

13 Show that $6 + \left[(x+5) \div \frac{x^2+3x-10}{x-1} \right]$ simplifies to $\frac{ax-b}{cx-d}$ where a, b, c and d are integers.

(Total for Question 13 is 4 marks)

15 Make m the subject of the formula $f = \dfrac{3m + 4}{m - 1}$

.....

(Total for Question 15 is 3 marks)

17 There are some small cubes and some large cubes in a bag.
The cubes are red or the cubes are yellow.

The ratio of the number of small cubes to the number of large cubes is 4 : 7

The ratio of the number of red cubes to the number of yellow cubes is 3 : 5

(*a*) Explain why the least possible number of cubes in the bag is 88

.....

.....

.....

(1)

All the small cubes are yellow.

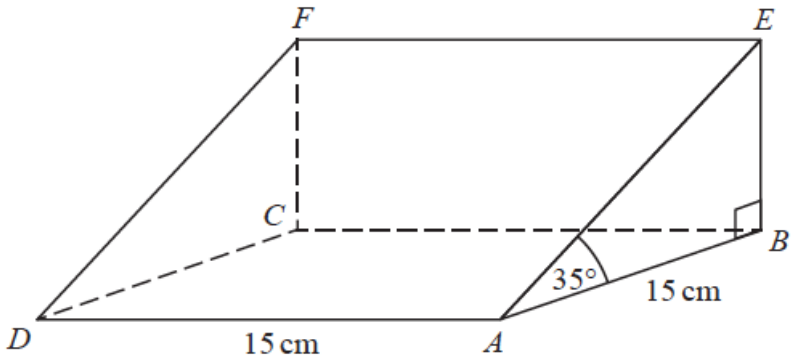
(*b*) Work out the least possible number of large yellow cubes in the bag.

.....

(3)

(Total for Question 17 is 4 marks)

19 The diagram shows a triangular prism.



The base, *ABCD*, of the prism is a square of side length 15 cm.
Angle *ABE* and angle *CBE* are right angles.
Angle *EAB* = 35°

M is the point on *DA* such that

$$DM : MA = 2 : 3$$

Calculate the size of the angle between *EM* and the base of the prism.
Give your answer correct to 1 decimal place.

..... °

(Total for Question 19 is 4 marks)

TOTAL FOR PAPER IS 39 MARKS